

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions

Remote

- Built a resilient browser-automation backend for MCA filings with retries, session recovery, public-site API lookups, and state validation, reducing manual filing time by **85%**.
- Optimized an AI-assisted company-registration pipeline for name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across workflow preparation.
- Containerized services with Docker, deployed to Google Cloud Run, and automated releases through GitHub Actions, improving deployment velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built MRI segmentation tooling for **95 anatomical regions**, integrating model inference, preprocessing, and visualization into an existing medical-imaging application.
- Reduced processing time by **87%** through async/parallel execution and fewer repeated passes across large medical volumes.
- Contributed production-facing changes under open-source review constraints, compatibility concerns, and careful debugging across existing workflows.

Open Source

[Contributions](#) 

- Contributed patches across Invesalious, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, working inside maintainer review processes for reliability, security, and integration quality.

Projects

Postificus | *Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, Prometheus*

[GitHub](#) 

- Built browser-automation workers around dynamic third-party publishing flows where sessions, page timing, external UI changes, and partial failures had to be handled explicitly.
- Designed queue-backed execution with Redis/PostgreSQL state, retries, DLQs, circuit breakers, and failure telemetry to make automation bugs inspectable and recoverable.

Seaweed Contest Platform | *Go, PostgreSQL, Redis, Judge0, AWS, Docker, Kubernetes*

[GitHub](#) 

- Built backend services for registration, code submissions, sandboxed Judge0 execution, scoring, and live contest rankings across **500+ concurrent users**.
- Designed async Go dispatch, verdict polling, PostgreSQL result schemas, Redis-backed runtime state, and Docker/Kubernetes deployment on AWS, reaching **sub-5s P95 verdict latency**.

Murdoc | *Python, FastAPI, MCP, OpenTelemetry, Prometheus, Docker*

[GitHub](#) 

- Instrumented an AI gateway with OpenTelemetry traces, Prometheus metrics, route profiles, health checks, and audit records tying model/tool requests to policy outcomes.
- Built replayable decision records for prompt, tool, retrieval, and output layers so high-risk AI behavior could be correlated with request context and failure modes.

Technical Skills

Languages: Go, Python, TypeScript, C++

Frameworks: Echo, FastAPI, Gin, Node.js

Tools: PostgreSQL, Redis, RabbitMQ, Docker, Kubernetes, AWS, GCP, Prometheus, OpenTelemetry, GitHub Actions